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SUHI analysis using Local Climate Zones—A comparison of 50 cities

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A B S T R A C T

Assessment of surface urban heat islands (SUHI) has been hampered by the lack of a consistent framework to permit consistent interpretation between cities. Local Climate Zones (LCZ) are a universal description of local scale landscape types based on expected variation at neighbourhood scale ($\geq 1\text{km}^2$) in and around cities. In this study, we investigate the suitability of the LCZ scheme for SUHI studies based on 50 cities from across the globe. For comparability we use an annual temperature cycle model for MODIS land surface temperature (LST) at different overpass times and multi-year mean Landsat 8 LST. The SUHI analysis shows significant differences in the intra-urban estimate of SUHI for different built LCZ types. Substantial variability of SUHI within LCZ classes and between cities exists and SUHI patterns vary by time of day. Landsat derived estimates have very high correlations to those from MODIS at a similar time. The use of an LCZ approach combined with annual SUHI estimates provides a promising approach for a consistent and comprehensive SUHI analysis framework subject to further work to assess the spatial scale of matching LST and LCZ data, filter for topographic effects, and include the phenological status.

1. Introduction

The urban heat island (UHI) is the most studied urban climate effect. It describes the influence of urbanization on the surface, sub-surface and air temperatures in cities and is revealed generally as higher values when compared with the surrounding landscape (Voogt and Oke, 2003). There are four types of UHI (surface, sub-surface, canopy layer and boundary layer) each of which is generated by different dominant physical processes and exhibit distinct temporal and spatial patterns although all are driven chiefly by the transformation of the natural landscape to a corrugated, mostly manufactured and less vegetated surface (Oke et al., 2017). The urban surface may be described as a composition of individual facets that are distinguished by their geometry (slope and aspect), distinct radiative and thermal properties and mostly dry state. The most commonly studied UHIs are the canopy layer (CUHI) and surface (SUHI) heat islands, which differ fundamentally in their energetic basic and temporal characteristics (Oke et al., 2017). While both are coupled and show relevant correlations (Nichol et al., 2009), the primary processes responsible for each differ (Mirzaei and Haghighat, 2010; Voogt and Oke, 2003).

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The CUHI is the result of the sensible heat fluxes at the facets of the urban canopy, that is the walls and ground below roof level and air exchanges with the atmosphere above roof level. Fixed and mobile observations using standard screen-level thermometers have shown that the CUHI is strongest at night under calm conditions and clear skies and that the magnitude of the urban effect increases from the urban-rural edge to the city centre (Arnfield, 2003; Mirzaei and Haghighat, 2010). Typical isotherm maps of the maximum CUHI reveal a pattern that closely matches the physical character of the urban landscape with the peak at the centre; hence the 'island' analogy (Kim and Baik, 2005; Saaroni et al., 2000). Within the urban canopy, significant variations in measured air temperature respond to local scale landscape changes (~1 km^2).

The SUHI, by comparison, results from modifications of the surface energy balance at urban facets, which alters the available net radiation and directs it towards sensible heat exchanges with the overlying air (via convection) and underlying substrate (via conduction) (Clinton and Gong, 2013; Voogt and Oke, 2003). The resulting temperature of the surface depends on its geometric relationship to the solar beam, its albedo and conductivity, exposure to airflow among others (Morrison et al., 2018; Voogt and Oke, 2003). Consequently, the pattern and magnitude of the SUHI is extremely heterogeneous in space and time and is difficult to measure in a meaningful manner (Coutts et al., 2016; Schwarz et al., 2011).

The most common assessment of the SUHI uses satellite platforms to measure the land surface temperature (LST), which implicates relevant limitations. In particular, remote sensing provides a partial view of the urban surface (Coutts et al., 2016; Keramitsoglou et al., 2012; Sobrino et al., 2012; Tomlinson et al., 2011). For example, the SUHI as viewed from a satellite platform is dominated by the areal fractions of ground cover and roofs of different shapes (the walls are mostly invisible) and while the ground may be shadowed, roof surfaces, which have anomalous thermal and radiative properties, are usually illuminated (Morrison et al., 2018; Voogt and Oke, 2003). As a result the satellite-based SUHI shows extreme spatial variation with highest magnitude during the daytime (Allen et al., 2017; Coutts et al., 2016; Hu et al., 2016; Jiang et al., 2018; Keramitsoglou et al., 2011; Tran et al., 2006). Furthermore, spaceborne remote sensing involves a trade-off between spatial and temporal resolutions (Sismanidis et al., 2015). Currently none of the available sensors (Tomlinson et al., 2011) can fulfil the requirements to observe the urban surface temperature in adequate resolution (Bechtel et al., 2012). Moreover, LST acquisition is limited and exhibits a 'noisiness' associated with the specific atmospheric and surface conditions at observation time. This substantially limits the representativeness of individual LST acquisitions (Göttsche and Olesen, 2001; Göttsche and Olesen, 2009) and subsequently their validity for SUHI analysis. On the other hand comparisons of cities are aggravated by the limited availability of spaceborne data, since different cities cannot be observed under the same or at least comparable conditions (Schwarz et al., 2011). Most SUHI studies are based on a few scenes, and thus representativeness is very limited (Bechtel and Sismanidis, 2018). A solution is the use of long time series to assess the annual average SUHI (Zhou et al., 2013) thereby suppressing noise resulting from diurnal variability. Peng et al. (2012) conducted a comprehensive study using five years of MODIS data for 419 cities and Zhang et al. (2010) used 3 years of MODIS data to analyse > 3000 settlements globally. Clinton and Gong (2013) also used annual MODIS data and studied the global location of SUHIs and their controls, while Schwarz et al. (2011) and Zhou et al. (2013) focused only on European cities. Most recently, Chakraborty and Lee (2019) computed SUHI from MODIS for over 9500 urban clusters using the Google Earth Engine cloud processing platform and also investigated trends as well as large scale climatic and seasonal variations. An alternative is the use of LST time series to model the annual temperature cycle (ATC) (Bechtel, 2012; Fu and Weng, 2015) that captures seasonal variation and partly compensates for different data availability throughout the year. Few SUHI case studies have used this approach: for example Bechtel (2015a) for Dutch and Belgium cities; Kaloustian et al. (2017) for Beirut, Lebanon; Ching et al. (2018) for Sao Paulo, Brazil and Mumbai, India; Wiesner et al. (2018) for Hamburg; and Bechtel (2015b) for cities on five continents. However, most of these only investigated the magnitude of the SUHI and not the spatial differentiation that results from the character of the built-up landscape.

Historically, scientific advances in both the CUHI and SUHI have been hampered by the lack of a landscape framework that would allow the results to be interpreted consistently. In the absence of such a framework, researchers have used a variety of landuse and landcover datasets. In SUHI studies various parameters such as vegetation indices, impervious surface fraction, night lights, urban extent, cluster size, and population were used as descriptors instead (Clinton and Gong, 2013; Imhoff et al., 2010; Zhang et al., 2010; Zhou et al., 2013); in CUHI studies the discrimination could be as simple as urban/non-urban or use of differing classifications that makes comparisons difficult (Stewart and Oke, 2012). Stewart (2011) found that much of the scientific knowledge on CUHI was of little general value owing to a lack of metadata particularly with regard to the location of the observation sites. Critically, the magnitude of the UHI effect depends on both the character of the urban landscapes and those of the non-urban landscapes against which the effect is measured (Hawkins et al., 2004; Mirzaei and Haghighat, 2010; Schwarz et al., 2011). Local Climate Zones (LCZ) were developed as a suitable universal description of local scale landscape types (Stewart and Oke, 2012); in other words, they capture the variation in microclimates that typify neighbourhoods (≥1 km^2) in cities. The basic scheme accounts for 10 urban and 7 natural types and each is associated with a range of values for distinguishing parameters that directly influence 2-m air temperature in the canopy layer (e.g. impermeable surface cover, vegetative fraction, etc.).

The LCZ scheme has proved to be useful in planning CUHI studies and interpreting the results. Many studies have shown the relevance of the LCZ types to CUHI observations (Alexander and Mills, 2014; Fenner et al., 2017; Fenner et al., 2014; Lehnert et al., 2015; Skarbit et al., 2017; Stewart et al., 2014) but its general suitability for SUHI studies still has to be investigated. Yet, it seems plausible, that the scheme may be useful for SUHI studies of surface temperature variability at local scale. Both are closely coupled via the surface energy balance (Morrison et al., 2018; Oke et al., 2017) and some of the factors that define the LCZs are also important controls on surface temperature (i.e. surface cover fractions and materials). Thus we hypothesize, that the LCZ scheme is a relevant discriminator of surface temperature at the neighbourhood scale (even though much higher variability at the facet scale can be expected). So far a number of studies have applied the LCZ scheme to investigate SUHI for individual cities using different sources of thermal data, for example: Landsat 8 for Delhi National Capital Region (Budhiraja et al., 2017), Wuhan, China (Wang et al., 2017), Fuzhou, China (Zhongli and Hanqiu, 2016), Taipei, Taiwan (Shih, 2017), Las Vegas and Phoenix, USA (Wang et al., 2018). Others used both Landsat and ASTER data, e.g. for the Prague and Brno, Czech Republic (Geletic et al., 2017; Geletič et al., 2016) or the Yangtze River Delta, China (Cai et al., 2017) or Landsat and MODIS for Dubai, UEA (Nassar et al., 2016). Skarbit et al. (2015) used a low-cost small-format digital imaging system in early night hours in Szeged, Hungary and found that different LCZ classes have different surface temperature characteristics. Koc et al. (2018) used high-resolution airborne remote sensing data to evaluate a GIS based LCZ classification method. The aforementioned works suggest that regardless of the city studied or the satellite data employed, the LST differ significantly among the LCZ (some inconsistencies can be caused however, by the pronounced morphology of urban areas. In general, LCZ representing densely built-up areas and heavy industries showed the highest LST, while those representing suburban and natural land covers made up the coolest zones (Geletič et al., 2016). These studies also report distinct diurnal (daytime/ nighttime) and seasonal differences in LST intensity and suggest that each LCZ has a very distinct thermal response (Nassar et al., 2016).

However, most of these studies used individual to a few LST acquisitions to characterize the SUHI which implies a certain degree of uncertainty and hinders comparison between cities. Therefore, a comprehensive comparison applying the LCZ concept to SUHI for multiple cities is missing. To that end, we conduct an initial comparison of SUHIs based on long LST time series and the LCZ scheme for 50 cities globally. The objectives are to investigate the LCZ scheme's suitability in general and more specifically the use of WUDAPT L0 data for SUHI assessment, and based on that the differences in the SUHI related to the built environment and natural surroundings. Additionally, a first comparison of SUHIs of multiple cities shall be conducted based on LCZ and LST time series. The long term goal is to integrate these methods into a consistent and comprehensive SUHI analysis framework.

2. Data and methods

2.1. Local climate zones

The LCZ scheme is used in the World Urban Data Access Portal Tools (WUDAPT) project that was developed to meet the need for better information on the form and function of cities globally (Ching et al., 2018). WUDAPT collects, stores, and disseminates these data at different levels, each of which represents a degree of spatial precision on the elements of the urban landscape. The WUDAPT level zero (L0) product is a raster map of a city and its surrounding area, which has been categorised into LCZ types (Stewart and Oke, 2012). Each type along with the associated building, cover and material properties are shown in Fig. 1. These properties can serve as inputs to high resolution urban climate models.

The WUDAPT L0 product is generated using freely available Landsat imagery, crowdsourced training areas from the community, and the open source SAGA software (Bechtel et al., 2015). Community contributions are used to generate LCZ maps which are

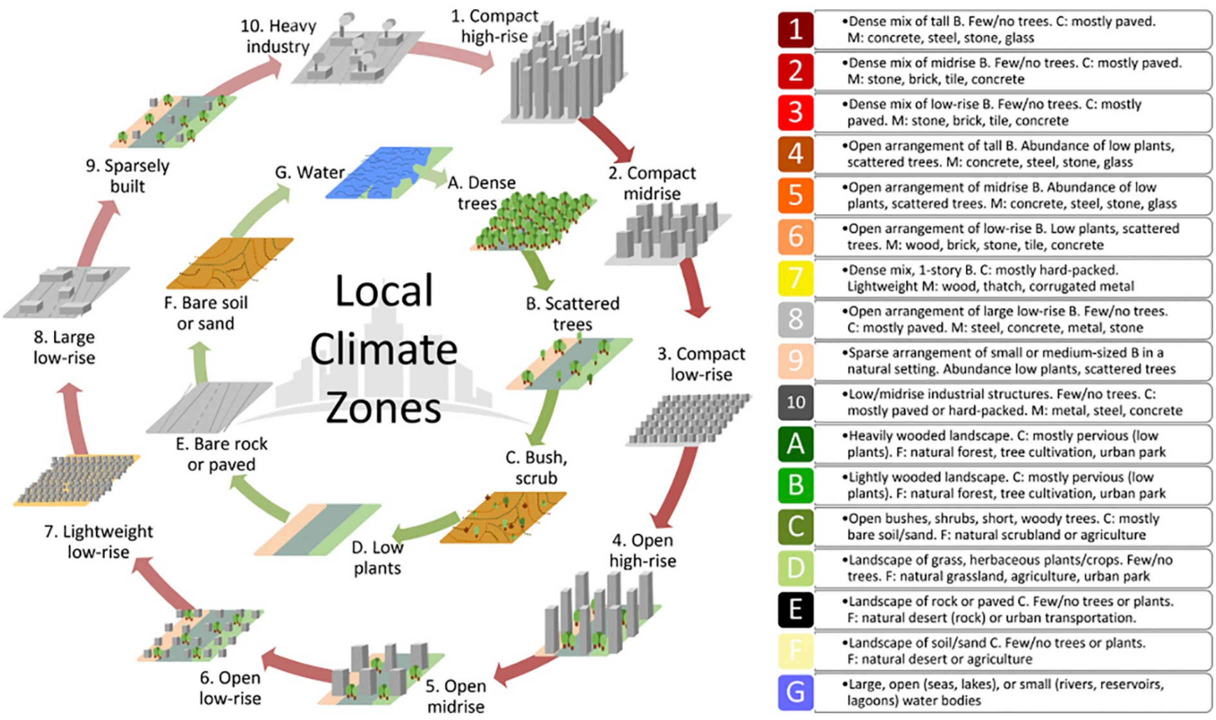

Fig. 1. Urban (1 – 10) and natural (A-G) LCZ types and their characteristics and colour code used in the WUDAPT framework. B: Buildings; C: cover; M: materials; F: function; Tall: > 10 stories, Mid-rise: 3–9 stories, Low: 1–3 stories (Bechtel et al., 2017), adopted from (Stewart and Oke, 2012).

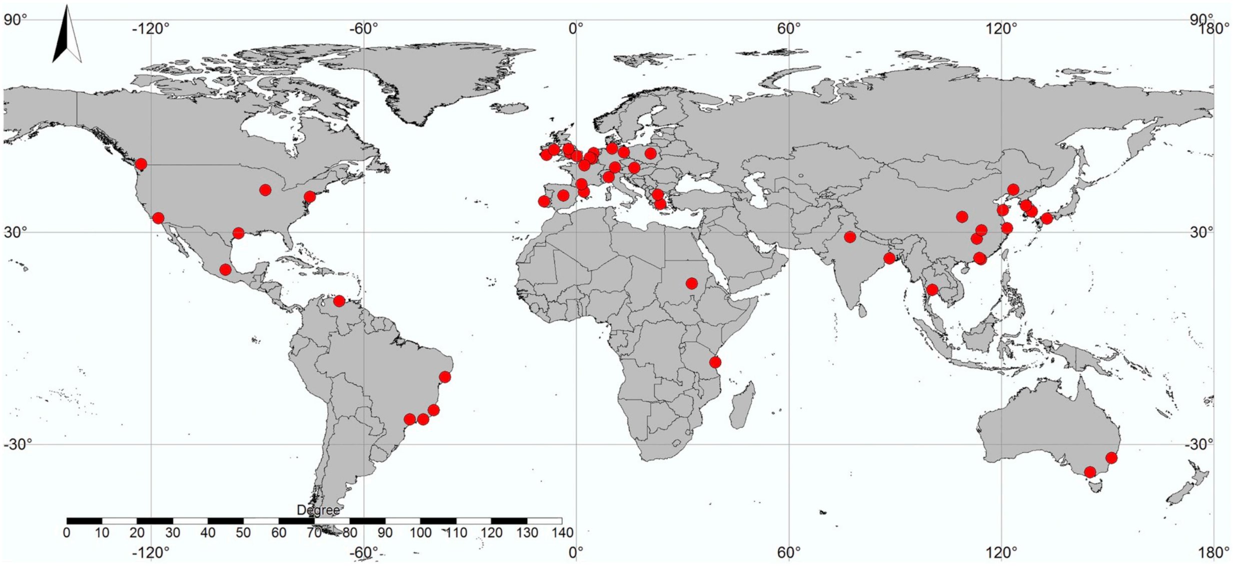

Fig. 2. Geographic distribution of the cities considered.

subsequently evaluated for quality and added to a database (Bechtel et al., 2019). Fig. 2 shows the geographic distribution of the considered cities.

For this study, cities with comparably good classification accuracies were selected from the WUDAPT database and manually checked; a few cities were excluded due to complex topography or shorelines. The final list of cities as well as relevant metadata can be found in Table 1. The indicated accuracy values are calculated following a 25-fold cross-validation for that city (Kaloustian and Bechtel, 2016). The overall accuracy (OA) indicates the percentage of correctly classified pixels while the weighted accuracy (WA) takes the class similarity into account (Bechtel et al., 2017). The mean OA for all considered cities is 0.79 for all and 0.66 for the urban LCZ types only, the mean WA is 0.94. Although the cities were selected with respect to their accuracies, it has to be kept in mind, that they still contain errors.

The standard grid resolution of the LCZ maps is 100 m. For this study the L0 (LCZ) maps that were derived using a majority filter of 3 pixels radius (300 m) were used. Subsequently, for each map a class wise morphological opening filter with a disk shape structuring element of 5 pixels radius was applied, removing borders and indifferent regions from the map. The effect is demonstrated for the case of Hamburg, Germany, in Fig. 3. The resulting map only includes fairly homogenous regions (i.e. pure LCZs), of at least about 0.9 km diameter.

2.2. Annual land surface temperature from MODIS

To account for the large temporal variability of LST, a simple model of the annual temperature cycle (ATC) was applied to a LST time series of several years (Bechtel, 2012). This reduces a large and noisy dataset to one described by a small number of ATC parameters, that capture the thermal response of the surface. These parameters in particular include: the mean annual surface temperature (MAST); the yearly amplitude of surface temperature (YAST), and a phase shift (θ) compared to the variation in solar irradiation. For this study, the ATC data were taken from a dataset derived from MODIS LST data for 2003–2014 at 1 km^2 spatial resolution and reported in (Bechtel, 2015c). All three parameters were re-projected to the target grids of the 50 LCZ maps using B-Spline interpolation in SAGA (Conrad et al., 2015), but subsequently only MAST was evaluated. Thus this study focuses on the annual mean SUHI mostly neglecting the seasonal changes. If a city was split on several MODIS tiles these were merged in an additional pre-processing step, but where only small natural areas distant from the city were missing this was accepted. Fig. 4 shows the example of Khartoum (Sudan); the four images show the LCZ cover and the resampled nighttime ATC parameters MAST, YAST and θ.

One of the major advantages of MODIS over other satellite products is that it has four overpass times (01:30, 10:30, 13:30, 22:30 local time), which provides a rich sample of the diurnal response of the Earth's surface. LCZ maps and LST for the four times of day for selected examples can be seen in Fig. 5. A main disadvantage however is the spatial resolution mismatch with the LCZ data. Thus Landsat 8 LST was used as a second data source.

2.3. Land surface temperature from Landsat

City-wide mean multi-year Landsat 8 LST maps were obtaining by implementing the single channel algorithm from Jimenez-Munoz et al. (2009; 2014) in Google Earth Engine (Gorelick et al., 2017). The procedure generally follows the workflow described in Parastatidis et al. (2017) and is used here to generate mean LST images based on all Landsat 8 cloud- and cloud shadow-free pixels

Table 1
Selected Cities from WUDAPT L0 database. Author, production date, overall accuracy (OA) and weighted accuracy (WA), latitude, longitude, Training data ID.

City	Author	Date	OA	WA	lat	lon	Training data
Amsterdam	Ran Wang, Benjamin Bechtel	2017-07-03-1534	0.80	0.95	52.365	4.861	Amsterdam_DFC_fin
Aracaju	Max Anjos	2017-02-02-1622	0.80	0.94	−10.893	−37.098	Aracaju_MaxAnjos_MF_20170129
Athens	Panagiotis Sismanidis	2017-08-31-1852	0.73	0.95	38.029	23.710	athens_PS_BB
Augsburg	Christoph Beck	2017-09-07-1014	0.80	0.95	48.337	10.895	augsburg(2016-14-10){Christoph_Beck}[[Benjamin_Bechtel]]_HUMINEXref
Bangkok	Torben Kraft	2017-09-01-1254	0.70	0.92	13.654	100.456	Bangkok(20170831){Torben_Kraft}[[BB]]
Barcelona	Joan Gilabert Mestre	2017-02-02-1726	0.78	0.95	41.487	2.133	Barcelona_Joan_Gilabert_Mestre_MF_20170129
Berlin	Daniel Fenner	2016–12-21-1448	0.68	0.92	52.535	13.361	Berlin_DanielFenner_BB_20151117_v6
Birmingham	Ines Friedrich	2017-09-22-1637	0.88	0.95	52.481	−1.952	Birmingham(2017-01-16){Ines_Friedrich}
Brussels	Marie-Leen Verdonck	2016-11-29-1305	0.62	0.90	50.854	4.360	[Brussels][16-12-2015][Verdonck]
Caracas	Karenia Cordova	2017-03-22-1227	0.81	0.96	10.474	−66.925	Caracas_KareniaCordova_MF_20170308
Changsha	Chao Ren	2017-03-22-1230	0.89	0.96	28.213	112.979	Changsha_ChaoRen_MF_20170309
Chicago	Micheal Foley	2017-07-03-1548	0.94	0.98	41.931	−87.875	Chicago_MichealFoley_20170701
Cork	Paul Alexander	2017-02-02-1745	0.84	0.96	51.887	−8.461	Cork_PaulAlexander_MF_20170130
Daegu	KSD	2017-06-15-1456	0.74	0.91	35.880	128.588	Daegu_KSD_NB_20170608
Dar_es_Salaam	{BB	2018-01-10-0258	0.87	0.97	−6.844	39.213	Daressalaam(2015-11-19)][KrugAlexander]{{BB}}
Dublin	Paul Alexander	2017-02-02-1747	0.88	0.96	53.342	−6.286	Dublin_PaulAlexander_MF_20170130
Ghent	Marie-Leen Verdonck	2017-09-08-1820	0.69	0.94	51.075	3.728	ghent(Marie-Leen_Verdonck}[[Benjamin_Bechtel]]_HUMINEXref
Hamburg	Michael Kottas	2017-11-09-1344	0.87	0.96	53.585	10.001	Hamburg_{M_Kottas}[[Benjamin_Bechtel]](2017-11-09)
Hong_Kong	Chao Ren	2016-11-16-2019	0.75	0.92	22.368	114.144	Hong_Kong_ChaoRen_MF_20151122
Houston	NialBuckley	2017-09-04-1018	0.82	0.94	29.686	−95.364	Huston{NialBuckley}[[MF]](20,170,214)
Khartoum	Micheal Foley	2017-03-22-1315	0.76	0.92	15.539	32.594	Khartoum_MichealFoley_200170320
Kolkata	Debashish Das	2017-06-15-1501	0.73	0.93	22.653	88.357	Kolkata_DebashishDas_MF_BB_20170607
Lisbon	Max Anjos	2017-02-27-1728	0.75	0.92	38.769	−9.181	Lisbon_Max_Anjos_MF_20170217
London	Nial Buckley	2017-06-12-2242	0.88	0.96	51.529	0.066	London_NialBuckly_TK_MF_20170612
Los_Angeles	Caroline Evans	2017-09-01-1151	0.52	0.85	33.993	−118.068	Los_Angeles(16062017){Caroline_Evans}[[Torben_Kraft,Benjamin_Bechtel]]
Madrid	Oscar Brousse	2016-11-16-2119	0.62	0.89	40.428	−3.727	Madrid_OscarBrousse_BB_20151122
Manchester	Micheal Foley	2017-02-27-1752	0.69	0.89	53.487	−2.278	Manchester_MichealFoley_20170217
Matsuyama	Deepak Thapa	2017-02-27-1756	0.99	1.00	33.846	132.763	Matsuyama_DeepakThapa_MF_20170220
Melbourne	Luke Gebert	2017-02-27-1807	0.81	0.93	−37.846	144.973	Melbourne_LukeGebert_MF_20170220
Mexico_City	Alvaro Lentz	2017-02-27-1824	0.74	0.93	19.457	−99.089	Mexico City_Alvaro_Lentz_MF_DW_Benjamin_Bechtel_20170220
Milan	Maria Brovelli	2016-11-16-2213	0.72	0.91	45.637	9.078	Milan_MariaBrovelli_MF_20151016
New_Delhi	Kshama Gupta	2017-09-04-1049	0.59	0.87	28.592	77.199	New_Delhi(23-02-2016){Kshama_Gupta}
Paris	Guillaume Dumas	2016-11-16-2237	0.87	0.96	48.878	2.186	Paris_GuillaumeDumas_MF_20161001
Pearl_River_Delta	Cai Meng, Wang Ran	2017-02-02-1802	0.95	0.99	22.689	113.689	Pearl_River_Delta_CaiMeng_WangRan_MF_20170115
Philadelphia	Caroline Evans	2017-08-31-1937	0.76	0.94	40.025	−75.213	Philadelphia13062017CarolineEvans
Qingdao	Chao Ren	2017-03-22-1337	0.93	0.98	36.239	120.304	Qingdao_ChaoRen_MF_20170228
Rio_de_Janeiro	Max Anjos	2017-03-22-1356	0.86	0.96	−22.847	−43.324	Rio_de_Janeiro_Max_Anjos_MF_20170228
Sao_Paulo	Maria De Fatima Andrade	2017-03-22-1416	0.80	0.94	−22.897	−47.039	Sao_Paulo_MariaDeFatimaAndrade_MF_200150930_bb20161128_MF20170228
Seoul	Je Woo	2017-03-22-1435	0.93	0.98	37.586	126.897	Seoul_JEWOO_MF_20170228
Shanghai	Chao Ren	2017-06-09-1112	0.96	0.99	31.181	121.490	Shanghai_ChaoRen_MF_20170607
Shenyang	Chao Ren	2017-03-22-1454	0.82	0.95	41.978	123.362	Shenyang_ChaoRen_MF_20170228
Sydney	Yueyi Feng	2016-11-24-2314	0.82	0.94	−33.842	151.006	Sydney_Yueyi_MF_20161124

(*continued on next page*)

Table 1 (*continued*)

City	Author	Date	OA	WA	lat	lon	Training data
Thessaloniki	Panagiotis Sismanidis	2017-09-04-1317	0.89	0.96	40.599	22.941	Thessaloniki(04-09-2017){Panagiotis_Sismanidis}[[Benjamin_Bechtel]]_v7
Toulouse	Julia Hidalgo	2017-07-03-1710	0.82	0.94	43.594	1.372	Toulouse_JHidalgo_MF_20170702
Vancouver	Micheal Foley	2017-03-22-1509	0.72	0.91	49.206	−122.954	Vancouver_MichealFoley_20170228
Vienna	Oscar Brousse	2016–12-12-1214	0.72	0.91	48.188	16.334	Vienna_17112015_KrisH_OscarB
Vitoria	Taciana Albuquerque	2016-11-17-0100	0.75	0.94	−20.287	−40.323	Vitoria_TacianaAlbuquerque_MF_20151201
Warsaw	Monika Tomaszewsk	2016-11-17-0107	0.80	0.95	52.223	20.984	Warsaw_MonikaTomaszewsk_MF_20160920
Wuhan	Chao Ren	2016–12-01-1844	0.79	0.94	30.591	114.319	Wuhan_ChaoRen_MF_20161129
Xi'an	Chao Ren	2017-02-28-1351	0.83	0.94	34.310	108.917	Xi'an_ChaoRen_MF_BB_20170227

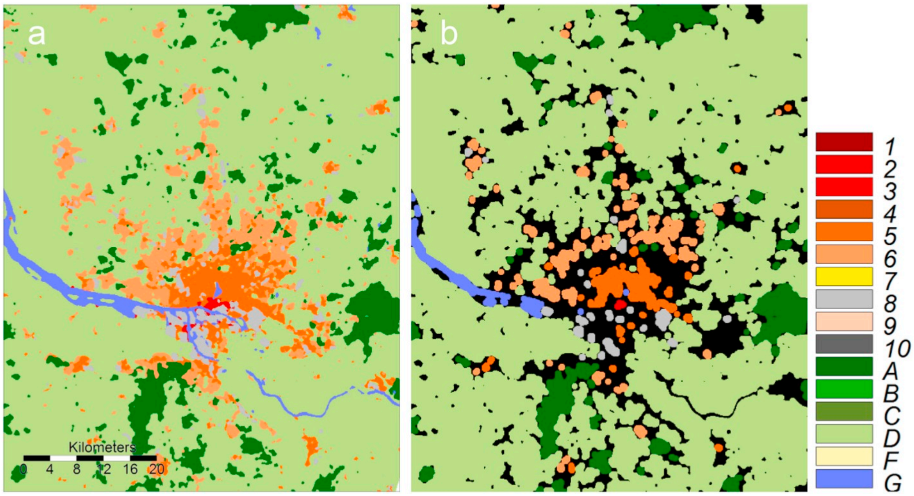

Fig. 3. LCZ map of Hamburg, Germany a) before and b) after class-wise morphological filtering. Black areas were not considered for further analysis.

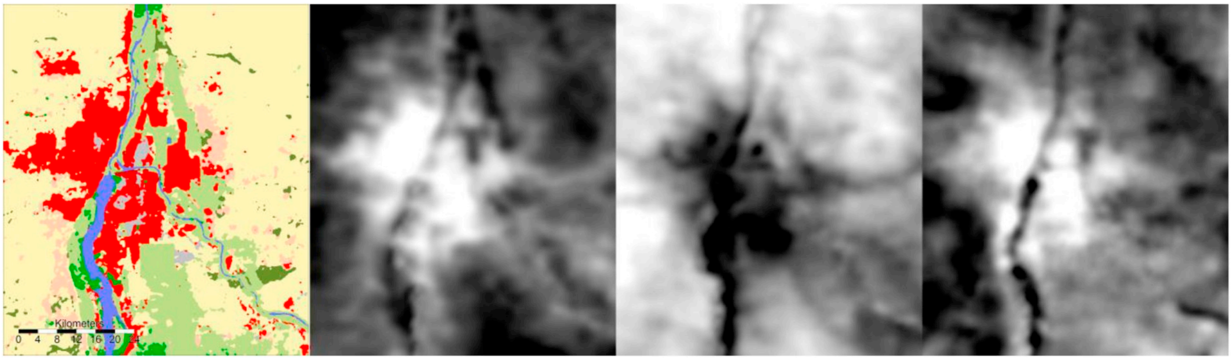

Fig. 4. Khartoum Local Climate Zones and annual temperature cycle parameters MAST, YAST, θ from MODIS at 1:30 local time.

between the years 2014 to 2017.

A list of input datasets is provided in Table 1. The Tier1 (T1) Landsat 8 datasets meet the geometric and radiometric quality requirements, in a variety of processing methods: at-sensor radiance, top-of-atmosphere (TOA) reflectance and surface reflectance (SR) (Table 1). These data have been atmospherically corrected using the Landsat 8 Surface Reflectance Code product (LaSRC) (USGS, 2018), providing a per-pixel quality assessment flags (pixel_qa band). For the current work, only clear pixels (that is, pixel_qa value of 0 for clouds and cloud shadows) were retained. Emissivity is NDVI-based (Jimenez-Munoz et al., 2009), as it captures phenological changes and their spatial pattern well, while alternative sources such as ASTER- or MODIS-derived emissivity are limited for areas with a dynamic land cover or small-scale heterogeneities (Parastatidis et al., 2017). Atmospheric correction of the thermal observations used total column water vapour from NCEP/NCAR reanalysis data (Kalnay et al., 1996), which is available on a 2.5 × 2.5° spatial resolution, every six hours (00, 06, 12 and 18 h UTC). Following Parastatidis et al. (2017), the average water vapour content for the two hours surrounding each Landsat 8 overpass was used. Finally, while all bands are available at a 30 × 30 m spatial resolution, the resulting LST image was produced at a 100 × 100 m resolution, in line with the original spatial resolution of the Landsat 8 thermal infrared bands and the default LCZ map target resolution (Bechtel et al., 2015).

Examples of Amsterdam (The Netherlands) and Changsha (China) are presented in Fig. 6. Compared to the MODIS annual LST data far greater spatial detail preserved by the higher resolution Landsat-8 TIRS sensor, but on the other hand significant artefacts can be seen that are a product of the Landsat orbits and resulting differences in local acquisition time.

The Landsat LST data were exported from GEE and subsequently interpolated to the LCZ grid using B-Spline interpolation in SAGA GIS (Conrad et al., 2015). The Landsat acquisition time is shortly before the first MODIS daytime acquisition and is thus labelled as 10:15 local time in the following. Key characteristics of all five LST datasets applied for SUHI assessment are summarized in Table 2.

An overview of the different LST datasets and overpass times is provided in Table 3. Due to differences in the processing, MODIS and Landsat LSTs are not directly comparable. In particular, they comprise different years, annual averaging, cloud clearing, sensor radiometry, and emissivity estimation. Yet the mean differences between MD-1030 and LS-1015 for all cities is only −1.7 K. Systematically lower temperatures in the MODIS data were expected due to sub-pixel cloud contamination in MODIS and a seasonal bias

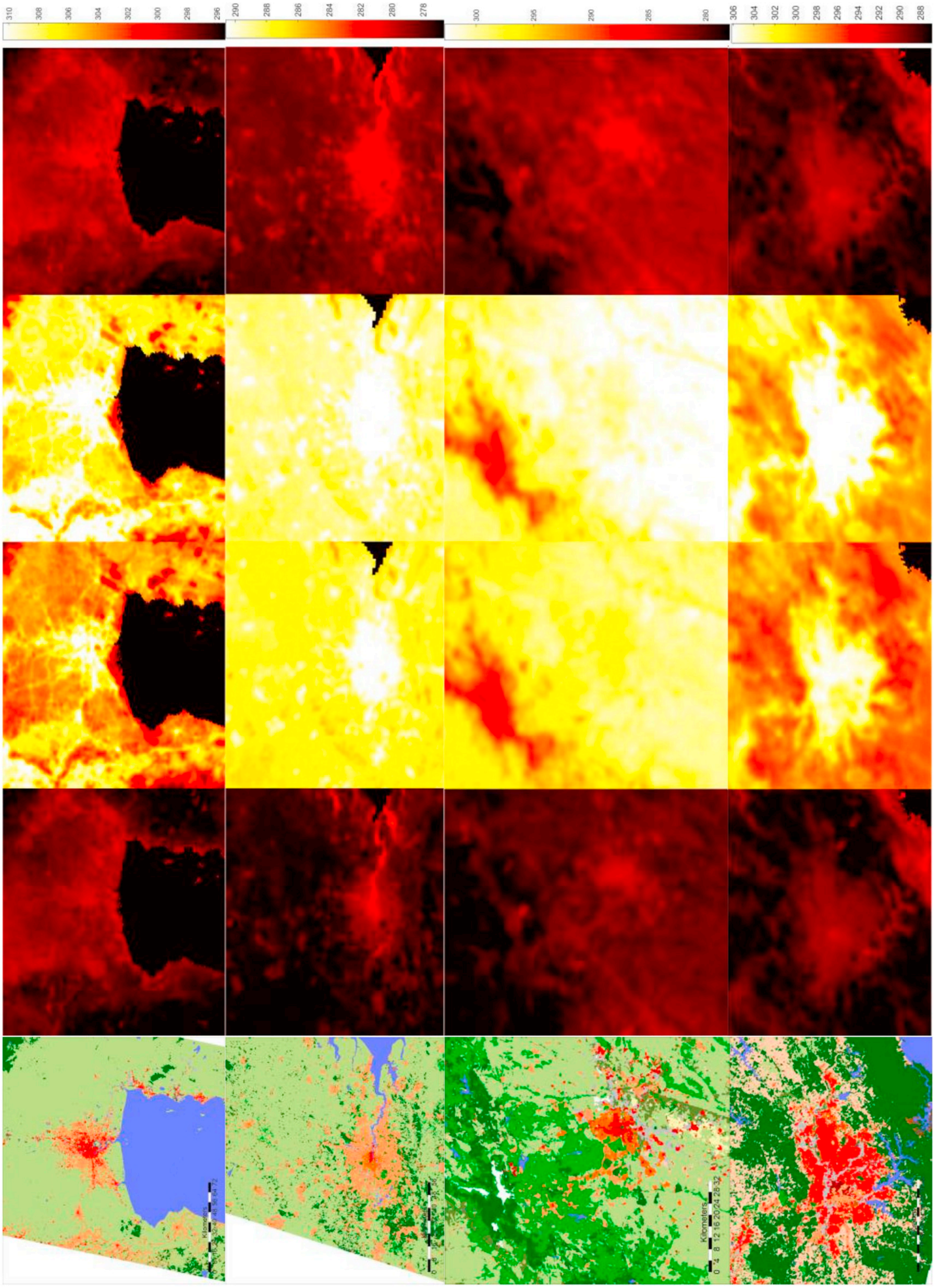

Fig. 5. LCZ (same colour code as in Fig. 2) and MAST for Bangkok (Thailand), London (UK), Madrid (Spain), and Sao Paulo (Brazil) (top to bottom) at 1:30, 10:30, 13:30, and 22:30 local time (from left to right) in K.

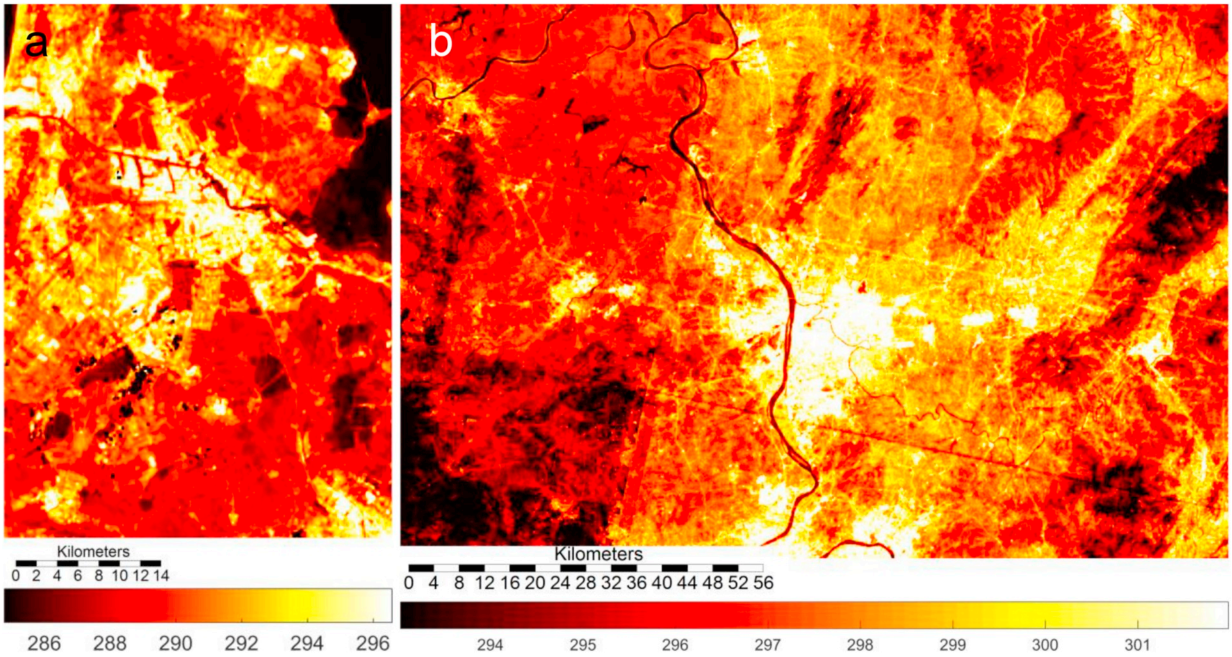

Fig. 6. LST from Landsat 8 in K for a) Amsterdam, Netherlands and b) Changsha, China.

Table 2
List of products in the GEE catalogue used to estimate the Landsat LST.

Input data	GEE Identifier
Landsat 8, Radiance at sensor from band 10	LANDSAT/LC08/C01/T1
Landsat 8, Brightness temperature from band 10	LANDSAT/LC08/C01/T1_TOA
Landsat 8, Surface Reflectance Product	LANDSAT/LC08/C01/T1_SR
NCEP/NCAR total column water	NCEP_RE/surface_wv

of the acquisitions towards summer (higher occurrence of clear sky conditions) in Landsat, while a certain spread can be expected by the selection of acquisition dates and years. However, not absolute temperatures but relative temperature differences are considered here. The applied data also limits the influence of thermal anisotropy, since for Landsat the observation angle is approximately nadir and for MODIS the observation angles vary constantly every 16 days. Since the ATC model involves data from all angles, the thermal anisotropy effect is considerably reduced (Hu et al., 2016; Jiang et al., 2018).

2.4. SUHI analysis

The comparison was conducted on the resolution, extent, and grid definition of the WUDAPT L0 LCZ maps. To minimize the impact of the effective resolution mismatch between the WUDAPT L0 LCZ and the MODIS annual LST data, spatial filtering was applied and only homogenous LCZs with a diameter of about 1 km were considered in the analysis (see 2.1 and Fig. 3). While their positions do not match the MODIS grid, it can be assumed that for the remaining regions the associated MODIS pixels are considerably or dominantly influenced by the respective LCZ types. For consistency the same filtered LCZs were used in the analysis with the Landsat LST. It has to be noted that individual Landsat LST datasets were used in the generation of the LCZ maps and thus these

Table 3
Key characteristics of applied LST datasets applied for SUHI assessment.

ID	Satellite/sensor	Local overpass time (HH:MM)	Processing
MD-0130	Aqua/MODIS	01:30	Mean Annual surface temperature (MAST) from annual temperature cycle model using 12 years of data. Product from (Bechtel, 2015c). 1 km spatial resolution.
MD-1030	Terra/MODIS	10:30	
MD-1330	Aqua/MODIS	13:30	
MD-2230	Terra/MODIS	22:30	
LS-1015	Landsat 8/TIRS	10:15	Mean LST of all cloud-free pixels within 4 years. Single Band (10), manual atmospheric and emissivity correction in Google Earth Engine. 100 m spatial resolution.

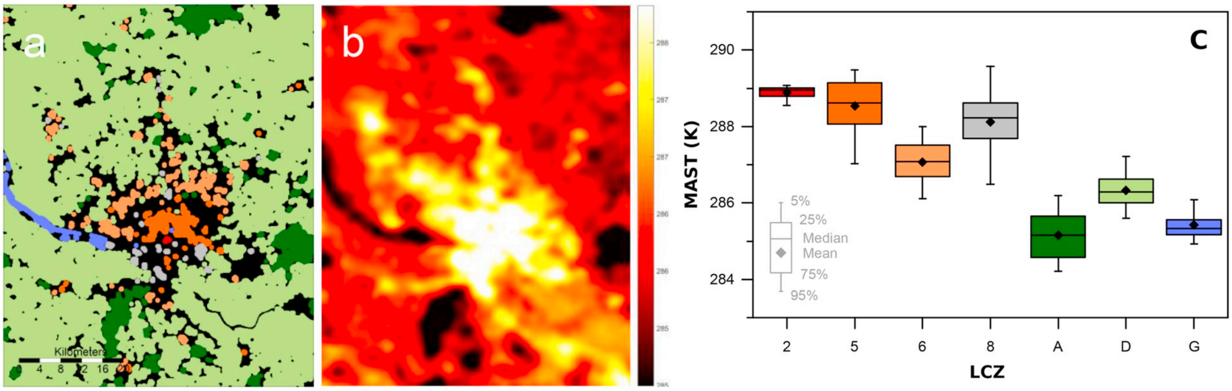

Fig. 7. a) LCZ (same colour code as in Fig. 2) and b) Mean annual surface temperature (MAST) at 22:30 for Hamburg, Germany. c) resulting distribution of MAST for different LCZ types.

two datasets are not fully independent. However, it was decided to use them in this analysis as they are, since first the LST is produced from a much larger number of Landsat images and second the aim of the study is to investigate the use of WUDAPT L0 data in SUHI studies. For each city the MAST from the four MODIS times as well as the mean LST from Landsat 8 were extracted for all remaining pixels as shown in Fig. 7 for the example of Hamburg, Germany.

Although the analysis was limited to 'pure' LCZ pixels, the assigned MODIS LST values for two adjacent Landsat pixels are highly related and cannot be considered as independent samples. Therefore, only the mean value per class and city was subsequently considered and all statistical tests were conducted considering the 50 cities as the sample.

Here the SUHI intensity is defined as the difference between the LST value of a LCZ type when compared with LCZ D (low plants), that is, $SUHI_{LCZx} = LST_{LCZx}\text{-}LST_{LCZD}$ (Stewart and Oke, 2012). Since LST of LCZ D shows spatial variation as well (see Fig. 7 b and c), its mean value was defined as reference surface temperature per city and observation time. In the following: first, the differences of the mean SUHI per LCZ type are shown; second, the distribution of SUHI values over the 50 cities are displayed in boxplots and; finally the distributions of the SUHI values are tested for differences between the LCZ types using the two-sample Kolmogorov-Smirnov test with a significance level of $\alpha = 0.05$. This nonparametric hypothesis test evaluates the difference between the cumulative distribution functions of the two sample data vectors. Statistical analysis and testing were conducted in MATLAB.

3. Results

3.1. Day and night SUHI from MODIS

The results of the SUHI analysis for the 50 cities are displayed in Fig. 8; the SUHI intensity for LCZ D is zero, by definition. While there is considerable variation between cities, some general characteristics are visible.

ll cities show positive SUHI values for the built classes (LCZ types 1–10) and compact types (LCZ 1–3) and commercial/industrial types (LCZ 8 and LCZ 10) generally have stronger SUHI intensities than the open types (LCZ 4–5 and especially LCZ 6). LCZ 9 (scattered buildings) has a considerably lower SUHI intensity than all other built classes. The magnitude of the SUHI for the built types is larger during daytime than during nighttime and the differences between types tend to be amplified during the day. For the natural classes, nighttime and daytime SUHIs are distinct: during daytime the "SUHI" (difference in LST) for forest (LCZ A) and water (LCZ G) is negative, while at night water is warmer and forest exhibits minor differences of different signs (slightly positive on average). A few cities have patterns that do not match these general, see for example Madrid and Khartoum, which have a negative SUHI during the day but positive SUHI at night. For Madrid (Fig. 5) this response can be attributed to the southeast area of the city region, which is classified as LCZ D but for much of the year has bare soil cover which has a very different thermal response. This is a limitation of the current WUDAPT LCZ mapping, which provides a static map of land-cover and does not account for phenology and changes over the year, in particular is does not allow to change the LCZ type seasonally as suggested by (Stewart and Oke, 2012). Again, this highlights the importance of the land cover surrounding cities when assessing the SUHI.

The distribution of the mean SUHI per LCZ for the 50 cities is shown in Fig. 9. It essentially confirms both the characteristic differences between LCZ types and the considerable variation between cities. The variation is much larger during daytime than during nighttime. A particularly large spread is associated with LCZ E (bare rock or paved) for both day and night and LCZ 1 (compact high-rise) during daytime. LCZ E can refer to either manufactured or natural surfaces and has been shown to cause confusion among the human operators who derive the training areas used in the WUDAPT L0 procedure (Bechtel et al., 2018; Bechtel et al., 2016). LCZ 1 on the other hand showed positive SUHI values for some cities (e.g. Bangkok, Caracas, Los Angeles, and Sao Paulo) and negative SUHI values for others (e.g. Seoul and Sydney).

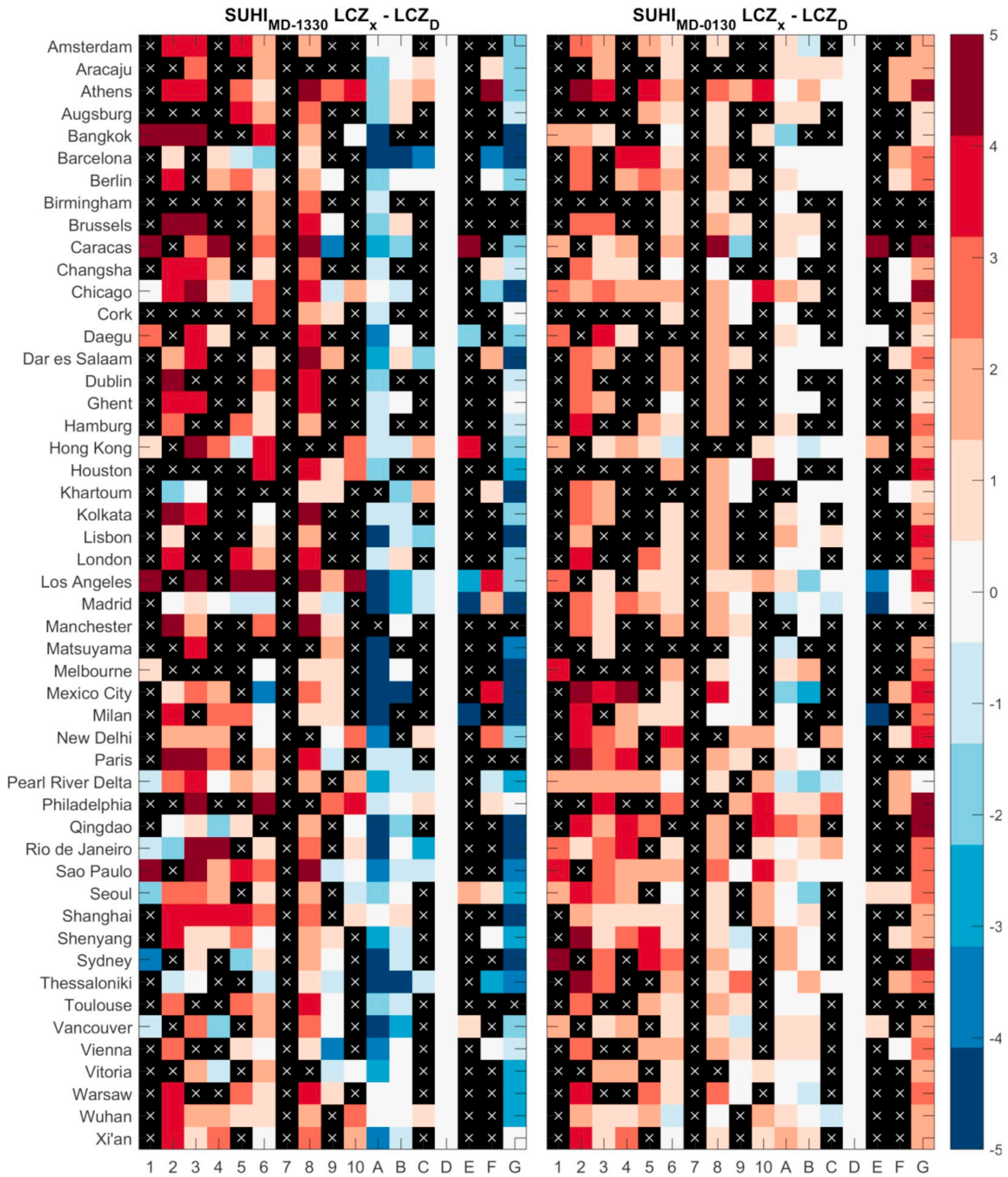

Fig. 8. Mean SUHI intensity in K per LCZ compared to LCZ D (low plants) for all cities at MODIS Aqua day- (MD-1330) and nighttime (MD-0130). Red colours indicate higher LST; blue colours indicate lower LST compared to LCZ D. Black squares with white x indicates that the LCZ is not present in the respective city. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Comparison between MODIS and Landsat SUHIs

The results of the Landsat SUHI and the most comparable MODIS overpass time at 10:30 are shown in Fig. 10. In general the results are consistent with similar SUHI magnitudes for LCZ types 1–6 in both datasets, which confirms the previous findings. However, the mean SUHI intensities for these types are slightly higher for the MODIS data (between 0.1 K for LCZ 6 to 0.9 K for LCZ 2). LCZ 1 shows much larger spread in MODIS data and is strongly influenced by outliers (the median difference is 1.9 K, see Fig. 10). Larger differences occur for the LCZ types 8 (1.9 K) and 10 (2.4 K), which show larger SUHI magnitude in the Landsat than in the MODIS data. Forested areas (LCZ types A and B) are relatively cooler in Landsat data (−1.6 and − 0.8 K respectively), LCZ E shows huge variation in both datasets, but is warmer in Landsat on average, while water (LCZ G) is cooler (−3.4 K). The higher positive and negative magnitudes in the Landsat data suggest higher purity of the data.

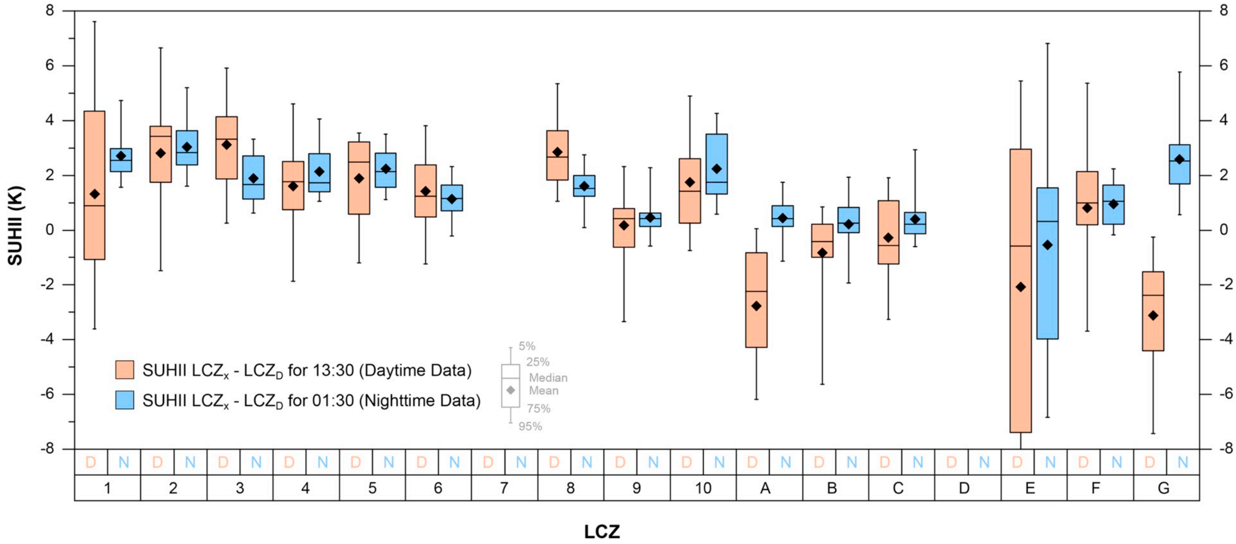

Fig. 9. Distribution of SUHI intensity in K for all cities and LCZs compared to LCZ D in K for MODIS Aqua day- and nighttime. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Comparison of acquisition times and sensors

The mean results for all cities and sensors are displayed in Fig. 11. The patterns are similar for the two MODIS daytime acquisitions (MD-1030, MD-1330) and the two nighttime acquisitions (MD-0130, MD-2230), while there are more pronounced differences between day- and nighttime. On average the warmest LCZ types are LCZ 2 (mean SUHI = 2.8 K), LCZ 3 (3.1 K), and LCZ 8 (2.8 K) at daytime (MD-1330) and LCZ 1 (2.7 K) and LCZ 2 (3.4 K) at nighttime (MD-0130). The average SUHI magnitude is remarkably similar between day- and nighttime (2.1 K, 1.8 K, 2.1 K, 2.2 K and 2.0 K for MD-0130, MD-1030, MD-1330, MD-2230, and LS-1015 respectively when considering the LCZs 1–6, 8 and 10), while the variation between cities is much larger during daytime (c.f. Fig. 9).

These mean values for all times and sensors essentially confirm the previous findings, i.e. similar results for Landsat and MODIS, but different patterns between day- and nighttime. Further support for this is provided by Pearson's linear correlation coefficients between the SUHIs (all cities and classes) between the times analysed (see Fig. 12). The correlation coefficients between the two daytime acquisitions (MD-1030, MD-1330) and two nighttime SUHIs (MD-0130, MD-2230) are very high (both $R = 0.98$), while those between day- and nighttime SUHIs are much lower (between 0.30 and 0.43). The correlation between Landsat and MD-1030 is 0.87.

Finally, the distributions of the SUHI intensities for all 50 cities were tested for differences. Fig. 13 shows the mean difference of the SUHI intensity per LCZ for MODIS 10:30 and 22:30 for all pairs with significantly different distributions at a significance level of

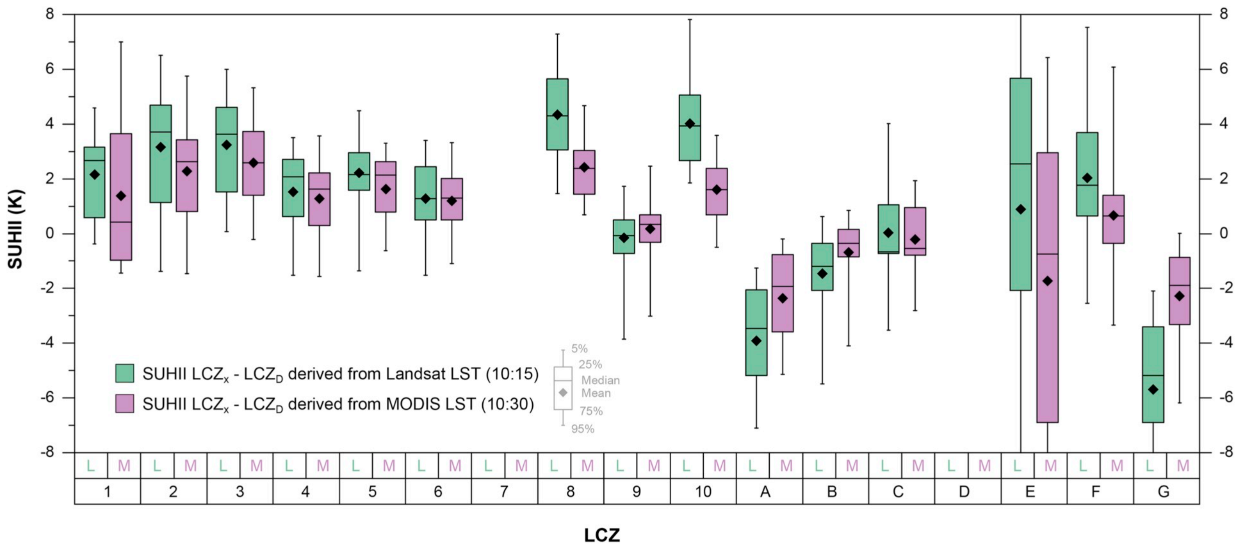

Fig. 10. Distribution of SUHI intensity in K for all cities and LCZ compared to LCZ D in K for Landsat LST, compared to MODIS Terra daytime.

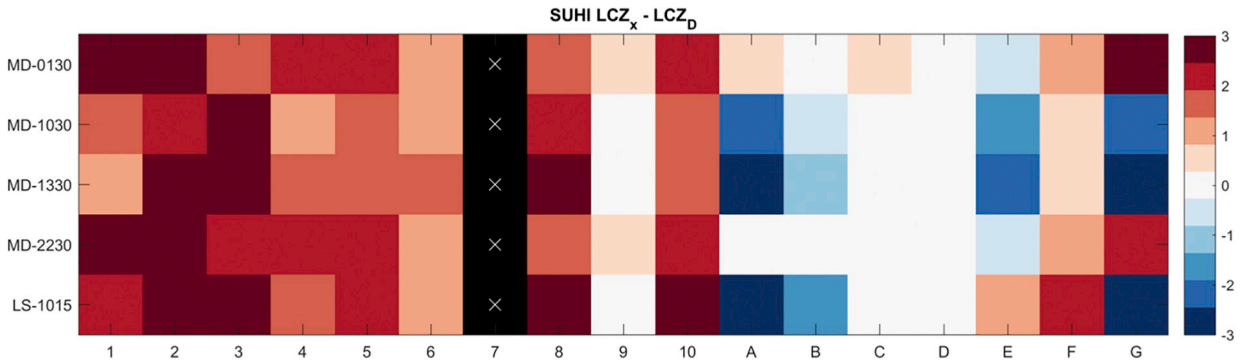

Fig. 11. Average SUHI intensity in K for all cities and LCZ compared to LCZ D for different acquisition times and sensors. Black with white x (LCZ 7) indicates that the LCZ type is not present in the study.

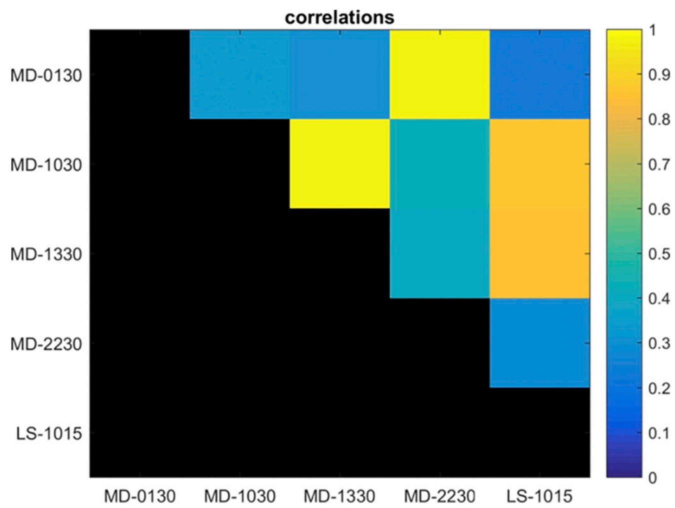

Fig. 12. Pearson's correlation coefficient between SUHI results of different acquisition times and sensors.

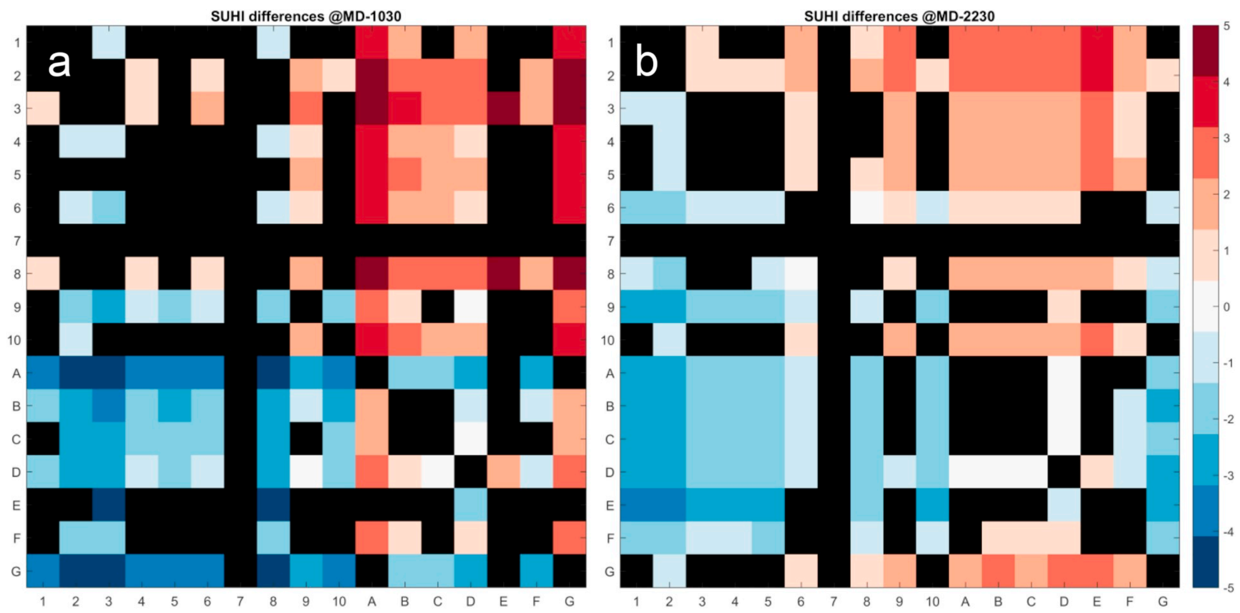

Fig. 13. Differences between SUHI intensity of different LCZs for 50 cities in K (row - column). Black indicates no significant differences at α = 0.05. a) MODIS 10:30; b) MODIS 22:30.

$\alpha = 0.05$ according to the Kolmogorov-Smirnov test. Generally, the natural classes except LCZ 9 differ from the urban classes. The sparsely built type (LCZ 9) differs significantly from all urban classes but only from the natural LCZ classes D and G (water) during nighttime; in terms of thermal surface charateristics LCZ 9 behaves closer to a natural, rather than urban type. Within the built types (except LCZ 9) LCZ 6 (open low-rise) is significantly cooler than all other built LCZ types during nighttime and than LCZs 2, 3, and 8 during daytime. Moreover, LCZ 2 and 3 are warmer than LCZ 4 during daytime and LCZ 2 is warmer than LCZs 3–5, 8 and 10 during nightime. However, these results somewhat differ for the Aqua acquisition times and additional tests suggested that the exact number of significantly different class pairs may also be sensitive to the period used in the time series analysis or the MODIS product version. Thus the results on different types should not be generalised but provide first evidence that certain differences exist, particularly between the compact (i.e. LCZs 2 and 3) and open (i.e. LCZ 6) types.

4. Discussion

These results in general can be considered as promising for further studies. Using a large dataset of 50 cities, the LCZ context detected SUHIs when the urban signal is compared to the rural (low plant) signal and revealed intra urban differences based on LCZ types. These magnitudes and signs of the SUHI are mostly in agreement with expectations with higher LST for the compact and commercial/industrial LCZ types. However, considerable variation in SUHI intensities were found between and within cities that deserve examination before the approach taken here can be considered appropriate for SUHI studies.

First of all there is a resolution mismatch between the annual LST data and the LCZ map. While we tried to limit this discrepancy by selecting ‘homogenous’ areas of $\geq 1\ \mathrm{km}^2$, some influence of the surrounding area, which dilutes the signal, must remain. There is some evidence of this in the results. In particular, the indifferent sign of the LCZ 1 SUHI may be related to this effect. The areas corresponding to LCZ 1 are usually only small patches of central business districts, which are often situated at waterfront areas (e.g. in Sydney, which shows the strongest daytime cool island for LCZ 1 with −3.6 K). Oppositely, in Los Angeles, which shows the strongest heat island for LCZ 1 (7.6 K), the LCZ 1 is embedded within commercial areas with LCZ 8 and LCZ 10. In comparison the Landsat data shows much smaller spread in the SUHI intensities of LCZ 1 for this class and in particular almost no negative values (except for Seoul and the Pearl River Delta). Yet also in the Landsat data, LCZ 1 exhibits lower LST than LCZs 2 and 3 during morning time. Similar effects were found in previous studies (Carnahan and Larson, 1990; Nichol, 2005). These may be related to a large thermal admittance favouring uptake of heat into storage at this time and shading of surfaces by the large canyon H/W during the morning hours (e.g. see the description in Peña (2008)) as well as from biases resulting from the satellite viewing geometry. Koc et al. (2018), using high resolution airborne thermal imagery in the early afternoon, also found LCZ1 to be consistently cooler at daytime. However, both the effect of adjacent LCZ zones and the relatively lower SUHI of LCZ 1 need more detailed investigation. Nevertheless the effective resolution of the LCZs lies between 100 s to 1000s m in horizontal scale (Bechtel et al., 2015; Stewart and Oke, 2012) and it is evident from this study that the MODIS annual LST is responding to surface features at a scale comparable to the LCZ (also in line with Zhou et al., 2013).

Second, the topography, which was not accounted for in this analysis, can be expected to have a relevant effect (Schwarz et al., 2011). This can be seen in some outliers in the SUHI, particularly in forest areas, which are often located in mountain ranges. This is for instance the case for Mexico City, which shows an extreme temperature difference of −13.2 K for LCZ A at MD-1330 because that LCZ type is located at higher altitude compared to the built up LCZ; similar results appear for Madrid (− 7.6 K), Vancouver (−6.2 K) and others. Likewise, the entire SUHI intensities can be affected if the reference class LCZ D is located in areas with higher altitude. This is the case in Los Angeles, where the natural areas are located in the San Gabriel Mountains north of the city. Accordingly, Los Angeles shows relatively high SUHI intensities for the built classes between 4.9 K for LCZ 10 and 7.8 K for LCZ 8. This underlines the importance of topographic correction or careful selection of reference areas in SUHI studies.

Third, the differences within the natural classes also stress the relevance of the surrounding land cover for SUHI studies, which is well in agreement with previous findings (Clinton and Gong, 2013; Imhoff et al., 2010; Schwarz et al., 2011; Zhou et al., 2014) and there is evidence that the SUHI signal of the coastal areas is influenced greatly by the water temperatures (Saaroni et al., 2000; Tran et al., 2006). A limitation of WUDAPT level 0 data scheme for natural land cover types is the absence of phenology and the systematic changes from plants to soil in agricultural areas in particular, since the level 0 data is static and also does not include the LCZ subscripts to describe specific surface properties, i.e. bare trees, snow, as well dry and wet soil (Stewart et al., 2014; Stewart and Oke, 2012). This can result in daytime cool islands in areas were LCZ D is dry soil over large parts of the year, which is in alignment with previous findings for semi-arid areas (Haashemi et al., 2016; Keramitsoglou et al., 2011; Rasul et al., 2016). Thus phenology and annual changes in surface cover and characteristics should be considered as relevant information in the WUDAPT level 1 and 2 databases. Problems are partially caused by the LCZ 9 class, which is unlikely to be homogenous at kilometre scale but as the built fraction occupies 20–30% of cover it is dominated by the thermal characteristics of natural cover (Bechtel et al., 2019; Bechtel et al., 2018). Nevertheless, using LCZ D as reference is a large improvement over SUHI studies that select the background (reference) LST based on distance from the city centre (e.g. Clinton and Gong, 2013; Zhou et al., 2013).

Moreover, there is a temporal mismatch between LCZ and the pre-computed MODIS ATC data, which was chosen to investigate the use of a readily available product (Bechtel, 2015c). This essentially implies that the cities are relatively unchanged at local spatial scale for about a decade, which is reasonable for most cities in Europe and North America but much less for the emerging cities in Asia, Africa, and South America. However, some of the most rapidly emerging cities (e.g. Pearl River Delta, Shanghai, and Shenyang) reveal comparably little difference between the MD-1030 and LS-1015 results for the urban classes. Yet for the future the use of more tailored ATC models should be considered. Moreover, the bias of the selected cities towards Europe (see Fig. 2) may influence the general finding, given that the surrounding biome is relevant for SUHI assessment (Imhoff et al., 2010). Thus more comprehensive studies using a larger set of cities on all continents are envisaged. Finally, it is expected that errors in the LCZ mapping resulting from dissimilar interpretations of the landscapes by different operators propagates to this analysis (Bechtel et al., 2017). To get an idea about the magnitude of this source of uncertainty, the mean SUHI magnitudes per class as in Fig. 11 were computed for each half of the cities with higher and lower OA separately. The median difference between both sets (for all LCZ types and LST datasets) was below 0.1 K but for some classes it was substantially higher (i.e. LCZ 1 is cooler for the more accurate half during daytime and LCZ E is much warmer for all acquisition times). However, these are related to the small sample size and outliers for these classes as discussed before. Thus these findings underline the need for a more comprehensive and regionally representative set of cities to extend the analysis.

The preliminary findings on the larger variation within daytime and nighttime SUHI are in agreement with previous studies (Clinton and Gong, 2013; Tran et al., 2006; Zhou et al., 2013) although they found slightly stronger daytime than nighttime SUHIs, e.g. an average of 1.5 K compared to 1.1 K for 419 cities in (Peng et al., 2012). However, the maximum surface urban heat and cool island intensities can occur at different times of day (Lai et al., 2018b) and months (Imhoff et al., 2010; Zhou et al., 2013), which is different from the temporal patterns of CUHIs. In particular, Zhou et al. (2013) report in their work an intra-annual variation of SUHIs and identified seven city cluster types in Europe, which exhibit regional separation; while Imhoff et al. (2010) showed that the ecological context is a statistically significant modulator of the diurnal and seasonal UHI's in the continental United States. The high correlations between MODIS and Landsat daytime SUHI compared to that of MODIS daytime and nighttime SUHI also suggest that the diurnal pattern is more relevant than the applied sensor, but likewise supports the value of coarse MODIS data in SUHI analysis at neighbourhood scale. To that end Coutts et al. (2016) also note the good agreement between the daytime SUHI retrieved from different sensors (in this case Landsat-7 ETM+ and very high resolution airborne LST). The strong agreement for both datasets, both of which have different caveats, strengthens the general findings. However, the results in Landsat data are generally more pronounced and the differences between both need more investigation, in particular in LCZ 1 and the stronger SUHI intensity of LCZ 8 and 10. So far it remains unclear, if the latter is related to emissivity or low thermal inertia, as suggested by Dousset et al. (2011) for Paris, France.

More generally, the differences between cities can be expected to be related to the macro-climate as well. We argue that by using a) long time series of LST data and b) a standardized description of built and natural land cover and land use types optimised according to their local scale thermal (atmospheric) impact, the presented methodology enables a standardized comparison between SUHIs and potentially their annual patterns when taking the full set of ATC parameters into account. However, towards the development of a standard SUHI analysis method a few additional challenges need to be addressed, including anisotropy (Krayenhoff and Voogt, 2016; Lagouarde et al., 2012; Voogt and Oke, 2003), which in model simulations has been shown to have significant variations between LCZs. These simulations (Krayenhoff and Voogt, 2016) show that urban morphology, more than material properties, exerts a strong control on the anisotropy with more built-up areas of the city with LCZs 1, 3 and 4 generating greater anisotropy than LCZ 6, in particular at higher solar elevation angle (lower latitudes). These estimates use very simple representations of the urban form. The addition of vegetation can further modulate the anisotropy, depending on the LCZ where increasing vegetation fractions tend to decrease anisotropy in compact geometries but increase it in open geometries (Dyce and Voogt, 2018).

Other factors to be considered include: emissivity of urban materials (Oke et al., 2017), seasonal variability especially of natural areas, temporal aggregation (Hu and Brunsell, 2013), quality control (Gawuc and Struzewska, 2016; Lai et al., 2018a) and accuracy of the LCZ maps as well as human interpretation of the scheme (Bechtel et al., 2017, 2019).

5. Conclusion

This study conducted a consistent and comprehensive SUHI analysis based on LST time series and the LCZ scheme for a large number of cities. More specifically, it presents a new approach for SUHI studies. Initial results from 50 cities were presented as proof of concept. Significant LST differences were found not only between built and natural classes but also within the built classes, which supports the suitability of the LCZ system and WUDAPT L0 data for SUHI analysis. Moreover, the use of multi-year time series or ATC models is strongly recommended for further SUHI studies. On the other hand there was substantial variation in the SUHI patterns between cities, which partly could be related to phenology, topography and mixed signals with adjacent classes. Thus, while some questions remain to be solved, we claim that this approach based on long time series of LST data and LCZ as surface description can be extended towards a comprehensive framework for SUHI analysis, which allows consistent investigation of SUHIs in cities of different sizes, compositions, and surrounding macro-climates.

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